# OIS-018 — Interoperability Specification

**Status:** Foundational

**Authority:** Implementation

**Depends on:**

- OIS-000 through OIS-017
- OAS-000 through OAS-031
- OCS-000 through OCS-032

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# Purpose

This specification defines the constitutional requirements for interoperability between independently developed Organodynamics implementations.

Interoperability is not protocol compatibility.

Interoperability is not identical data exchange.

Interoperability is the constitutional ability of independent implementations to cooperate while preserving equivalent understanding.

Implementations SHALL exchange constitutional meaning.

They SHALL not require identical engineering.

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# 1. Implementation Objective

The objective of the Interoperability Specification is to guarantee that independently developed implementations remain capable of constitutional cooperation throughout technological evolution.

Interoperability SHALL preserve:

Meaning.

Identity.

Historical continuity.

Architectural compatibility.

Operational independence.

Behavioral equivalence.

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# 2. Definition

Interoperability is the demonstrated constitutional capability of two or more independent implementations to exchange, interpret, and preserve equivalent constitutional understanding.

Operational diversity SHALL remain permissible.

Constitutional divergence SHALL not.

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# 3. Levels of Interoperability

Implementations MAY interoperate at multiple levels.

Examples include:

Identity exchange.

Event exchange.

Schema exchange.

Conversation exchange.

Knowledge exchange.

Decision exchange.

Federation exchange.

Future constitutional constructs SHALL integrate without redefining interoperability.

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# 4. Semantic Compatibility

Interoperability SHALL preserve semantic equivalence.

Equivalent constitutional meaning SHALL take precedence over identical operational representation.

Implementations MAY differ internally.

Observed constitutional understanding SHALL remain equivalent.

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# 5. Version Compatibility

Implementations SHALL explicitly declare supported versions of:

Constitutional Specifications.

Architectural Specifications.

Implementation Specifications.

Schema versions.

Compatibility SHALL remain observable.

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# 6. Exchange Integrity

Every constitutional exchange SHALL preserve:

Identity.

Relationships.

Historical attribution.

Ordering.

Architectural context.

Integrity SHALL remain independently verifiable.

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# 7. Operational Independence

Interoperability SHALL NOT require:

Shared infrastructure.

Shared databases.

Shared deployment.

Shared programming languages.

Shared cloud providers.

Independent implementations SHALL remain sovereign.

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## 8. Failure Tolerance

Temporary interoperability failure SHALL preserve:

Local operation.

Historical continuity.

Recoverability.

Future synchronization.

Failure SHALL not compromise constitutional integrity.

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## 9. Verification

Verification SHALL demonstrate:

Semantic equivalence.

Behavioral compatibility.

Recovery compatibility.

Historical preservation.

Federation compatibility.

Independent reconstructability.

## 10. Interoperability Test

The Interoperability Specification satisfies this specification when independently engineered implementations, developed without shared code or shared infrastructure, exchange constitutional constructs while preserving equivalent constitutional understanding and historical continuity.

## Conformance

An implementation conforms to this specification when constitutional interoperability remains demonstrable across independently developed systems regardless of implementation technology, operational architecture, deployment environment, or organizational ownership.

## Implementation Principle

Compatibility allows systems to communicate.

Interoperability allows civilizations to cooperate.

The purpose of Organodynamics interoperability is therefore not merely to exchange information,

but to preserve shared constitutional understanding across generations,

organizations,

and technologies that may never have existed at the same time.

**End of Specification OIS-018**